



MARTILIO BANZE

High-Performance Computing (HPC) Systems
Administrator
& Data Scientist

PERSONAL DATA

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HPC TOOLS

OpenHPC	● ● ● ● ●
Warewulf	● ● ● ● ●
SLURM	● ● ● ● ●
PBSPRO	● ● ● ● ●
Ganglia/Nagios	● ● ● ● ●
Docker/Singularity	● ● ● ● ●

OPERATING SYSTEMS

Linux/Unix	● ● ● ● ●
Windows	● ● ● ● ●

NETWORKING

TCP/IP	● ● ● ● ●
Infiniband	● ● ● ● ●

APPLICATION & WEB

Apache Tomcat	● ● ● ● ●
JBoss e WebLogic	● ● ● ● ●

PROGRAMMING & DATA SCI.

Anaconda/Python	● ● ● ● ●
QGIS	● ● ● ● ●
Oracle SQL	● ● ● ● ●
HTML e JavaScript	● ● ● ● ●
WRF Model	● ● ● ● ●
R/C++/Fortran/Matlab	● ● ● ● ●
M.O.Excel	● ● ● ● ●
Power B.I.	● ● ● ● ●

MARTILIO R. BANZE

HPC SYSTEMS ADMINISTRATOR & DATA SCIENTIST

PROFILE

HPC Ecosystems Administrator and Data Scientist, with extensive experience in the management, configuration, and optimization of HPC infrastructures. Specialist in SLURM, PBS Pro, MPI, and distributed computing, acting in cluster administration and resource optimization for intensive computing workloads. Concurrently applies Python in Machine Learning and Big Data analysis, developing scalable solutions for massive data processing.

EDUCATION

Bachelor's Degree in Meteorology 2013 - 2018

Eduardo Mondlane University (UEM) - Faculty of Sciences, Maputo

Bachelor's Degree in Physics and Mathematics Education 2018 - 2023

Pedagogical University (UPM), Faculty of Natural Sciences and Mathematics, Maputo

PROFESSIONAL EXPERIENCE

HPC Systems Administrator 2022-Current

MCTD/MoRENet, Maputo

HPC Tutor May 2021

UEM, Maputo

- Conducted an introductory training program in Astrophysics and Space Sciences (DARA Project), with an explicit structural focus on teaching Linux (Shell script) and Python.

Information Officer 2018 - 2021

National Institute for Disaster Management and Risk Reduction - INGD, Maputo

- Managed and maintained the central CENOE database, executing comprehensive tasks including advanced data analysis and data visualization pipelines.

Asset Manager 2016 - 2017

Movitel, SA, Matola

- Controlled, monitored, and audited technical assets, infrastructure hardware logs, and corporate office equipment at the Maputo Province level.

CERTIFICATIONS & TECHNICAL COURSES

Applied Data Science Lab 2026

WorldQuant University

ICT Government Staff Training March 2025

Huawei Mozambique

Data Science / ML & DL March 2025

Big Data Africa School, Cape Town - RSA

TCP/IP Architecture May 2024 – June 2024

Escola Superior de Redes, Brazil

Linux Administration June 2024 – July 2024

Escola Superior de Redes, Brazil

Practical Course in HPC Oct 2022 – Sept 2022

Center for High Performance Computing (CHPC), RSA

Advanced Network Management Mar 2022 – Mar 2022

INAGE, IP / MoRENet, Maputo

Astrophysics and Space Sciences 2018 – 2019

Hartebeesthoek Radio Astronomy Observatory (HartRAO), RSA

Linux (Bash Scripting) and Python 2018

Center for High Performance Computing (CHPC), RSA

LANGUAGES

- Portuguese
- English

INTERESTS

Systems & Applications Administration, High-Performance Computing (HPC), Data Science, and Climate Modeling.

QUALITIES

Commitment to excellence, demonstrating resilience in challenging environments. Professional ethics is a fundamental pillar, ensuring data and infrastructure integrity in all activities. Incessant focus on innovation, continuous improvement, and adaptive agility in dynamic environments.

CONFERENCES & WORKSHOPS

- 1) International Conference on HPC - SC25 (USA, St. Louis)

Nov 2025

Participated in the event, exploring advanced tutorials and prospective partnerships. Acted as an Invited Guest Speaker to present on MoRENet's HPC Infrastructure and its core contribution to national scientific research.
- 2) AI for Climate Services and Environmental Applications Workshop

Nov 2025

UEM in partnership with POLIMI
Intensive specialized training in Data Science, AI, Machine Learning (ML), and Deep Learning (DL) methodologies applied directly to climate datasets.
- 3) International Conference on HPC - SC23 (USA, Denver)

Nov 2023

Investigated technological breakthroughs, gaining sharp insights into upcoming industry trends and innovations. Acted as a Speaker in the pre-workshop session, addressing operational challenges faced in operationalizing HPC platforms in Mozambique.
- 4) RNP Forum 2023 (Brasília, Brazil)

Aug 2023

Attended the annual forum organized by the National Education and Research Network (RNP), a premier collaborative platform for high-level networking, relationships, and strategic updates on major emerging ICT trends.
- 5) 12th SADC Cyber-Infrastructure Experts Meeting & Annual HPC Meeting

Dec 2022

RSA, Pretoria
Provided a highly collaborative regional platform for exchanging technical knowledge, frameworks, and metrics in cybersecurity, alongside charting logistical breakthroughs and challenges in High-Performance Computing.
- 6) WRF (Weather Research and Forecasting) Model & GIS Training

Dec 2022

Advanced technical training program promoted by SADC, focused explicitly on utilizing high-performance compute architectures to accelerate and optimize national climate projections.

REFERENCES

Fialho Nehama, PhD Professor & Senior Researcher, UEM fialho.nehama@gmail.com	Elizabeth Leake, PhD Project Manager, Texas A&M University ltbeth2@gmail.com
Eng. Bryan Johnston, MSc Senior HPC Technologist II, CHPC, RSA Bjohnston@csir.co.za	Eng. Moisés Mucelo, MSc Academy Manager, INAGE, IP / MoRENet Moises.mucelo@morenet.ac.mz